IOE 611/ Math 663
Lecture 2: Convex sets (2)

Salar Fattahi

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Recap

- **affine combination** of x_1 and x_2: any point x of the form

$$x = \theta x_1 + (1-\theta)x_2 \qquad \text{where } \theta \in \mathbf{R}$$

  affine set: contains the line through any two distinct points in the set (i.e., a set that's closed under affine combinations)
  example: solution set of linear equations $\{x \mid Ax = b\}$

- **convex combination** of x_1 and x_2: any point x of the form

$$x = \theta x_1 + (1-\theta)x_2 \text{ where } 0 \leq \theta \leq 1$$

  convex set: contains line segment between any two points in the set (i.e., closed under convex combinations)

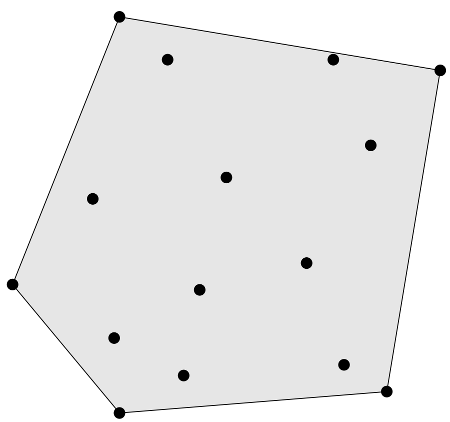
Convex combination and convex hull

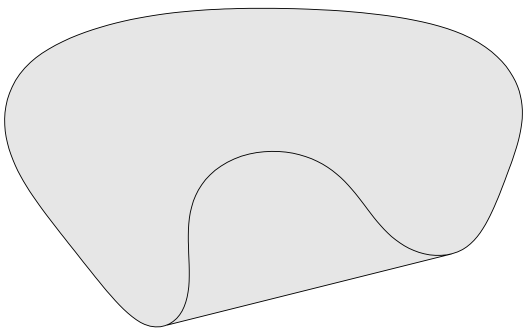
convex combination of $x_1, \ldots, x_k$: any point x of the form

$$x = \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_k x_k$$

with $\theta_1 + \cdots + \theta_k = 1$, $\theta_i \geq 0$

convex hull conv S: set of all convex combinations of points in S





Convex hull of any set S, is the smallest convex set that contain's S.

any other set D s.t $S \subseteq D$ & D is convex, then we must have

$$\text{conv}\, S \subseteq D$$

$$\Rightarrow \left.\begin{array}{l} S \subseteq D \\ D \text{ convex} \end{array}\right\} \Rightarrow \text{conv}\, S \subseteq D$$

Proof By contradiction:

① $S \subseteq D$, D is convex set But

$$\text{conv}\, S \not\subseteq D$$

That mean's there is at least one point that must exist's $x \in \text{conv } S$ in conv S such that $x \notin D$.

$\Rightarrow$ $\exists x \in \text{Conv } S$ s.t $x \notin D$

By the definition convex hull

$\exists x_1, x_2, \ldots, x_k \in S$ s.t

$x = \theta_1 x_1 + \theta_2 x_2 + \ldots + \theta_k x_k$ for some set of co-efficient's $\theta_1, \theta_2, \ldots, \theta_k$ s.t $\theta_1 + \theta_2 + \ldots + \theta_k = 1$ & $\theta_i \geq 0 \ \forall i$ & $x \notin D$.

$x \notin D$, But $x_1, x_2, \ldots, x_k \in D$ because $x_1, x_2, \ldots, x_k \in S$ & $S \subseteq D$

* But this contradict's convexity of D

Because we could find a set of co-eff-

ient's $\theta_1, \theta_2, \dots, \theta_k$ s.t $\sum \theta_i = 1$

& $\theta_i \geq 0 \ \forall i$, s.t their convex combination does not belong to D

$$\left.\begin{array}{l} x \notin D \\ x_1, x_2, \dots, x_k \in D \end{array}\right\} \Rightarrow \text{contradic's with convexity of } D$$

Convex cones

cone: a set C such that for any $x \in C$ and $\theta \geq 0$, $\theta x \in C$

A set C is a cone if for any $x \in C$

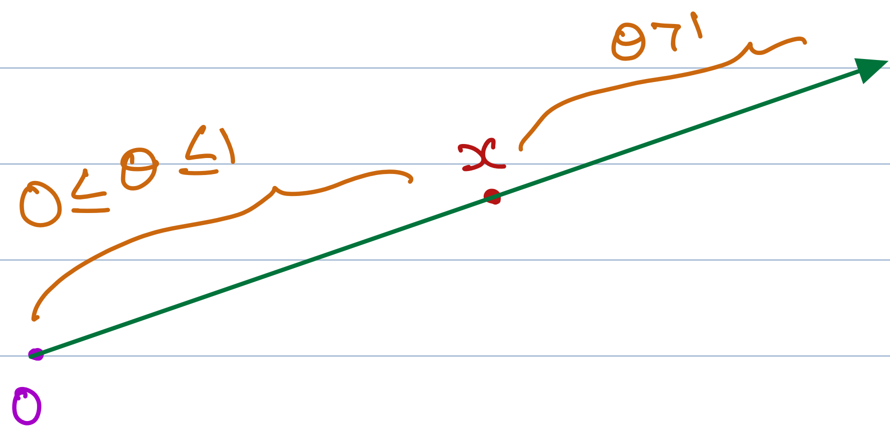
$\theta . x \in C$. x can be scalar, vector

matrix, But $\theta \in \mathbb{R}$ (scalar) & non

negative number $\Rightarrow$ $\theta \in \mathbb{R}^+$

$x \in C$, $\theta \geqslant 0$ $\Longrightarrow$ $\theta . x \in C$

intutively we can think of cone as set of Point's such that the ray that passes through every Point also belong's to set.



* A ray start's at origin & passes Point x. The set of all Point's on the ray is called cone.
* A cone should include all the ray's.

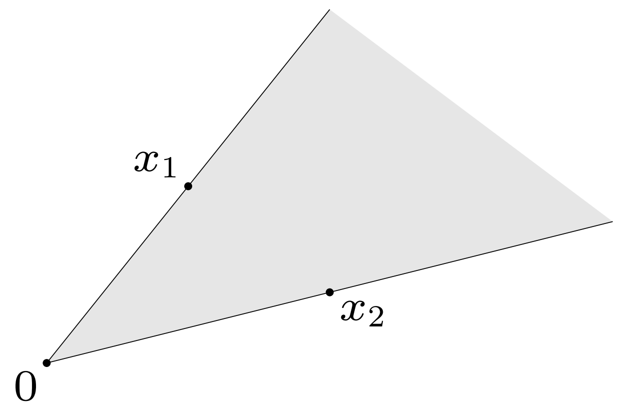
if we Pick any Point the ray start's at origin & passes the Point also belong's to cone.

Convex cones

cone: a set C such that for any $x \in C$ and $\theta \geq 0$, $\theta x \in C$

conic (nonnegative) combination of x_1 and x_2: any point of the form

$$x = \theta_1 x_1 + \theta_2 x_2 \text{ where } \theta_1, \theta_2 \geq 0$$

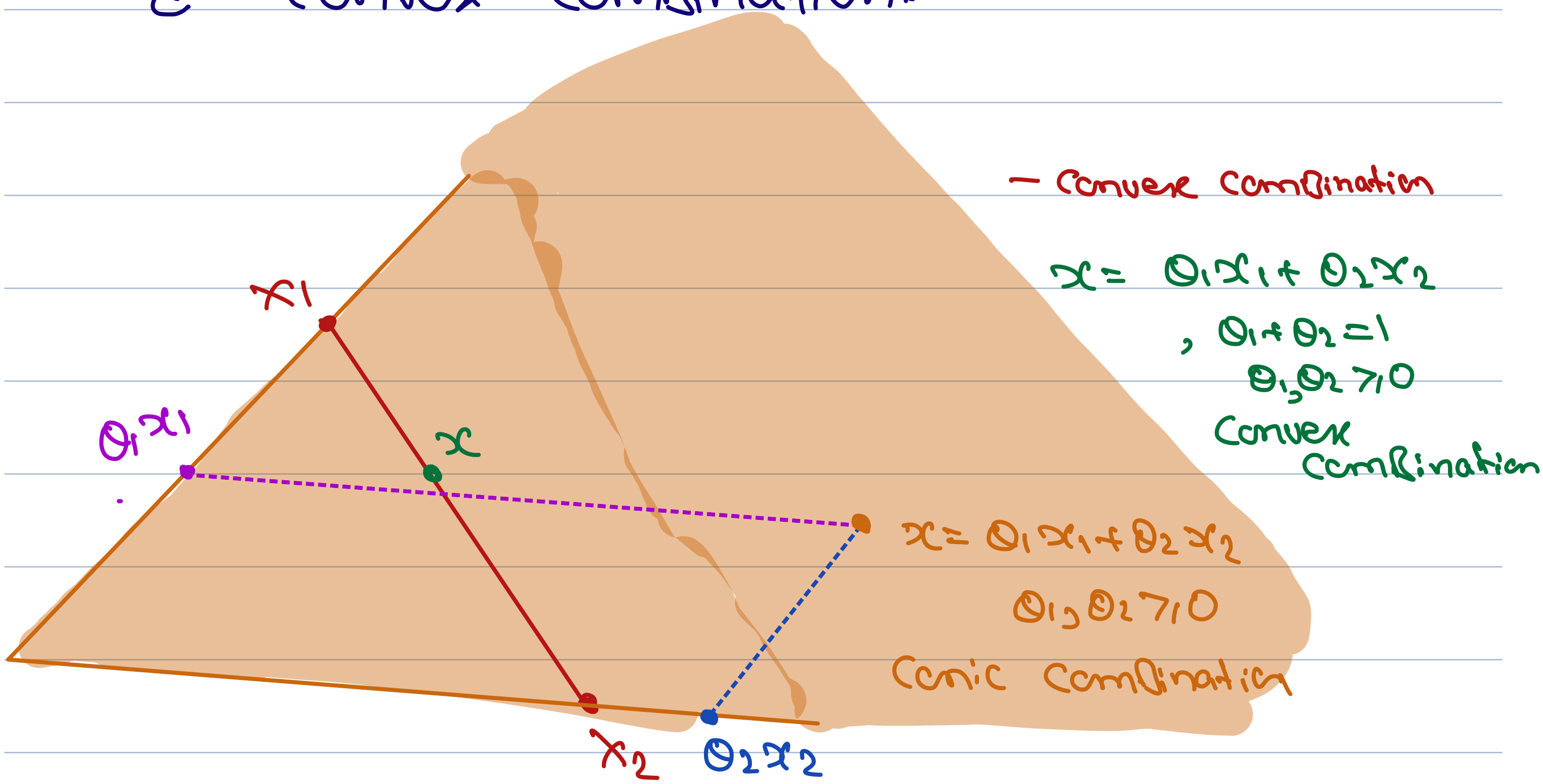


Conic Combination :-

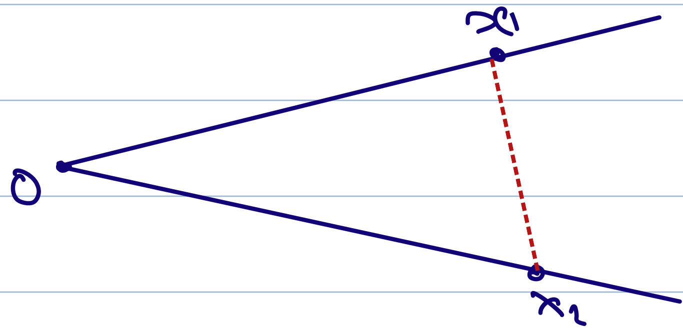
Given any two point's x_1 & $x_2 \in C$ their conic combination is any x that can be written as

$$x = \theta_1 x_1 + \theta_2 x_2 \quad \text{where } \theta_1, \theta_2 \geq 0$$

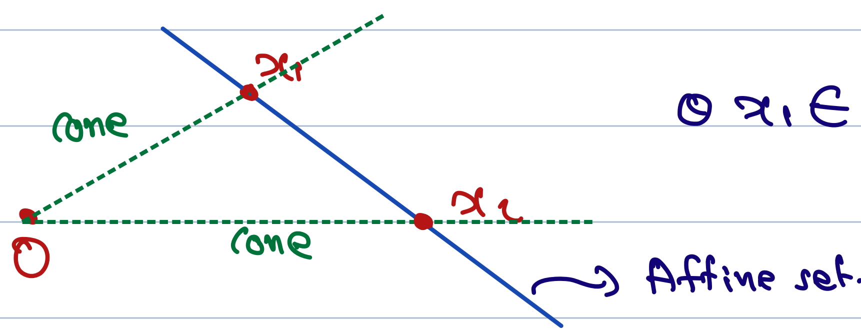
they do not need to sum up to 1.
that's the difference b/w conic combination & convex combination.



Are cones convex? NO



Are Affine set's cones? NO



$0 \; x_1 \in$ Affine set

if origin belong's to Affine set then it is cone.

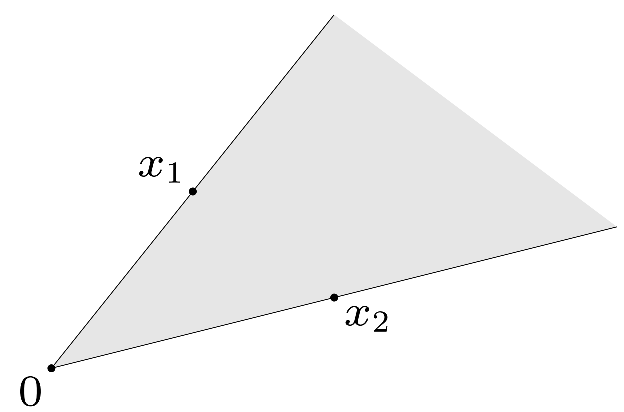
Convex set's are cone? NO

Convex cones

cone: a set C such that for any $x \in C$ and $\theta \geq 0$, $\theta x \in C$

conic (nonnegative) combination of x_1 and x_2: any point of the form

$$x = \theta_1 x_1 + \theta_2 x_2 \text{ where } \theta_1, \theta_2 \geq 0$$



convex cone: a cone that is convex, *i.e.*, set that contains all conic combinations

$$\{\theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_k x_k \mid \theta_i \geq 0\}$$

of points in the set (i.e., closed under conic combinations)

Cone's are not nessarly convex, But we can define convex cone's.

A cone that is also convex it is called convex cone.

turn's out A cone is convex iff

$$\Updownarrow$$

It contain's all of it's conic combination's

Proof: $(\Longleftarrow)$ if a set contain's all of its conic combination then it's convex

$$x_1, x_2 \in C \implies \theta_1 x_1 + \theta_2 x_2 \in C$$
where $\theta_1, \theta_2 \geq 0$

$$\Downarrow$$

$$\theta_1 x_1 + \theta_2 x_2 \in C$$
where $\theta_1, \theta_2 \geq 0$, $\theta_1 + \theta_2 = 1$

$$\Downarrow$$

convex set.

Examples

Affine / Convex / Cone

set that contain's only one element $C=\{x_0\}$

- Singleton

Affine ? ✓ Convex ? ✓ Cone ? ✗

only if $C=\{0\}$
(only Bounded set that is Cone is $C=\{0\}$)

- Line $C=\{x : a^Tx=b\}$

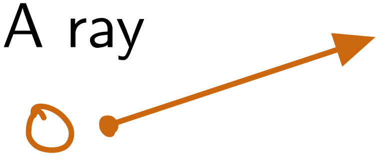
Affine ? ✓ Convex ? ✓ Cone ? ✗

(only if it passe's through origin, $b=0$)

- Line segment

Affine ? ✗ Convex ? ✓ Cone ? ✗

- A ray



Affine ? ✓ Convex ? ✓ Cone ? ✓

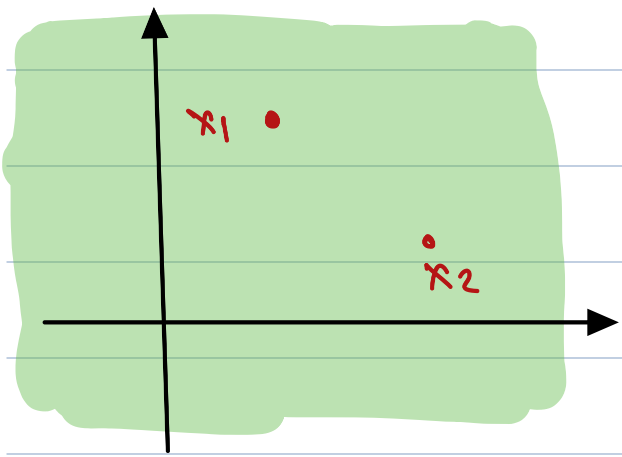
- A subspace if $x_1, x_2 \in C \Rightarrow \alpha x_1 + \beta x_2 \in C$

Affine ? ✓ Convex ? ✓ Cone ? ✓

SUMMARY

Let $x_1, x_2 \in \mathbb{R}^2$ be two independent Points

Linear Combination



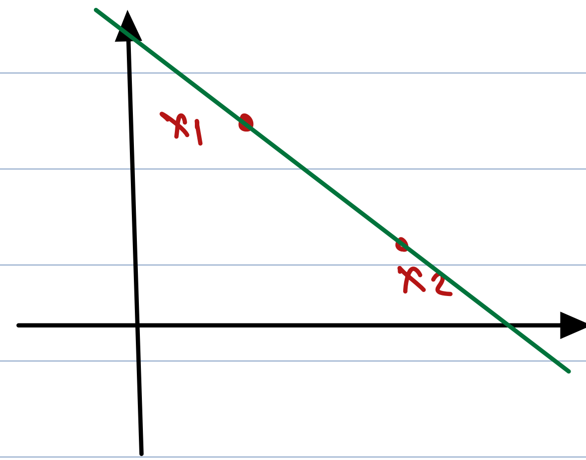
Linear Combination:

$$x = \theta_1 x_1 + \theta_2 x_2$$

$$\theta_1, \theta_2 \in \mathbb{R}$$

Entire $\mathbb{R}^2$

Affine Combination



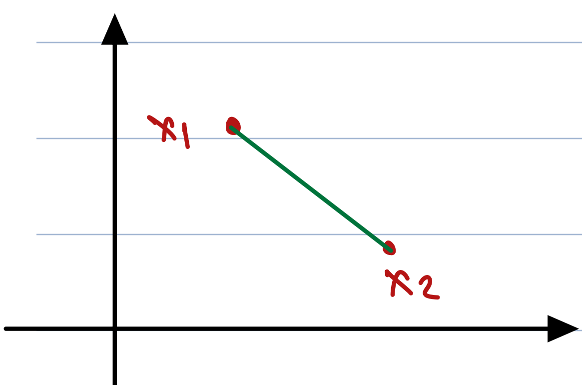
Affine Combination

$$x = \theta_1 x_1 + \theta_2 x_2$$

$$\theta_1 + \theta_2 = 1$$

Line passing through x_1, x_2

Convex Combination

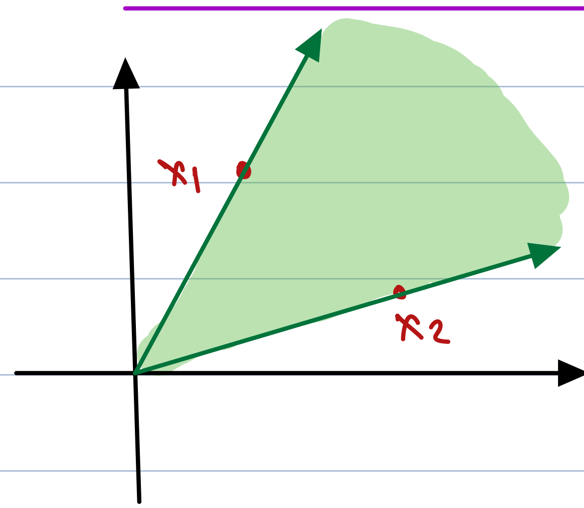


$$x = \theta_1 x_1 + \theta_2 x_2$$

$$\theta_1 + \theta_2 = 1 \; ; \; \theta_1, \theta_2 > 0$$

Line segment

Conic Combination



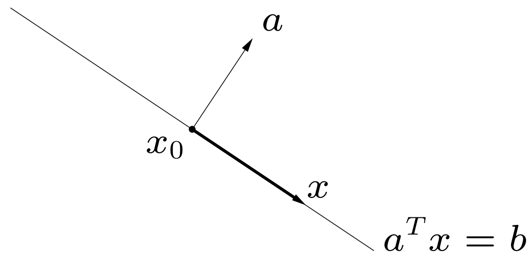
$$x = \theta_1 x_1 + \theta_2 x_2$$

$$\theta_1, \theta_2 \geq 0$$

Hyperplanes and halfspaces

$x \in \mathbb{R}^n$, $a \in \mathbb{R}^n$, $b \in \mathbb{R}$

hyperplane: set of the form $\{x \mid a^T x = b\}$ $(a \neq 0)$



- a is the **normal vector** of the hyperplane

A hyperplane is a set of form

$a^T x = b$, set of all x's s.t
$a^T x = b$ for a given vector a &
scalar b

$x \in \mathbb{R}^n$, $a \in \mathbb{R}^n$, $b \in \mathbb{R}$

Hyperplane's in 2D are line's.

without loss of generality we can assume

$\|a\| = 1$, why if it is $\neq 1$ we can divide
by $\|a\|$ on both side's.

a = normal vector of hyper plane.
Because any vector in this hyperplane
is orthogonal to that a.

set of all point's x whose inner
product with a is constant.

Another interpretation of hyper plane:

for any fixed $x_0 \in C$, we have

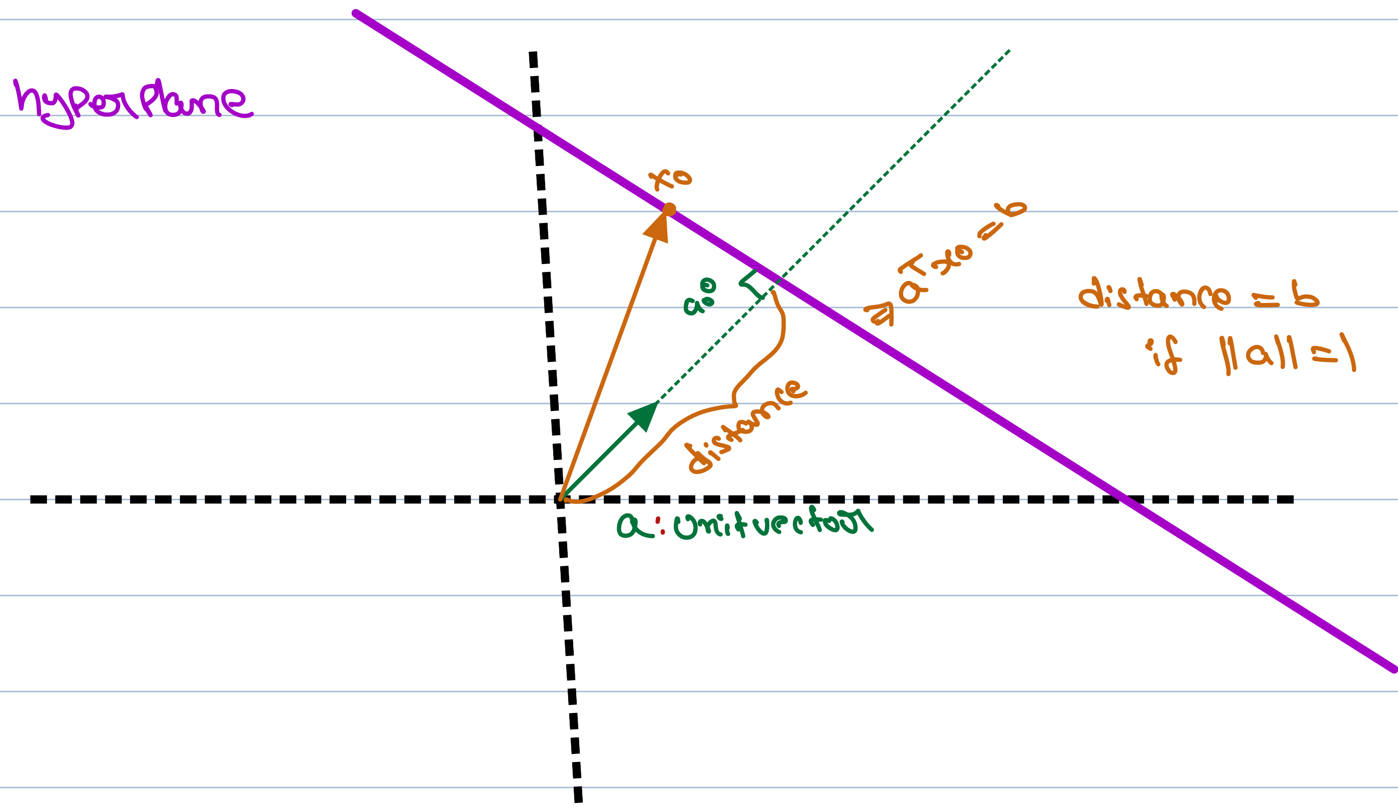
$$a^T(x - x_0) = 0 \quad \forall x \in C$$

$b \in \mathbb{R}$ determines the offset of the hyperplane from the origin.

$\Rightarrow \quad a^T(x - x_0) = 0 \qquad (a^T x_0 = b)$

$\Rightarrow \quad \{x \mid a^T(x - x_0) = 0\} = x_0 + a^\perp$

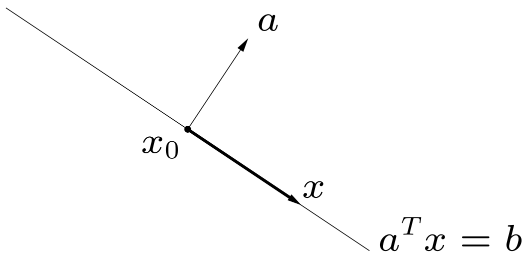
$a^\perp$ = orthogonal component of a
= set of all vectors orthogonal to a
$= \{v \mid a^T v = 0\}$



hyperplane
x_0
a
distance
a: unit vector
$\Rightarrow a^T x_0 = b$
distance $= b$
if $\|a\| = 1$

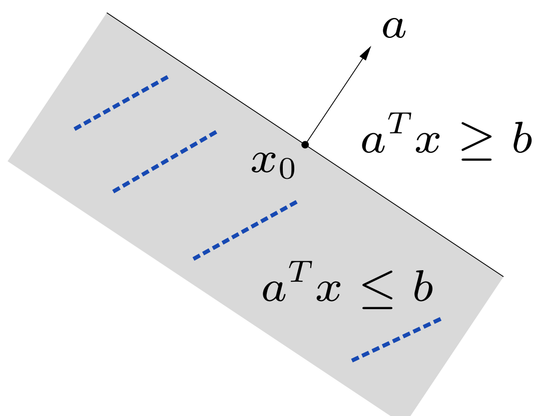
Hyperplanes and halfspaces

hyperplane: set of the form $\{x \mid a^T x = b\}$ $(a \neq 0)$



- a is the **normal vector** of the hyperplane

halfspace: set of the form $\{x \mid a^T x \leq b\}$ $(a \neq 0)$



for halfspace we replace equality with inequality in Hyper Plane

* when we have a hyperplane, we split the space into two part's, we Just Pick one of them.

Are hyper plane's convex?

$$a^T x = b \Rightarrow a^T(\theta x_1 + (1-\theta)x_2)$$
$$= \theta a^T x_1 + (1-\theta) a^T x_2$$
$$= \theta b + (1-\theta) b = b$$

convex.

Affine? yes

we discussed solution set of Bunch of equalities in a Affine set $\Rightarrow a^T x = b$ is one equality.

Are Halfspaces convex? Yes

Affine? NO

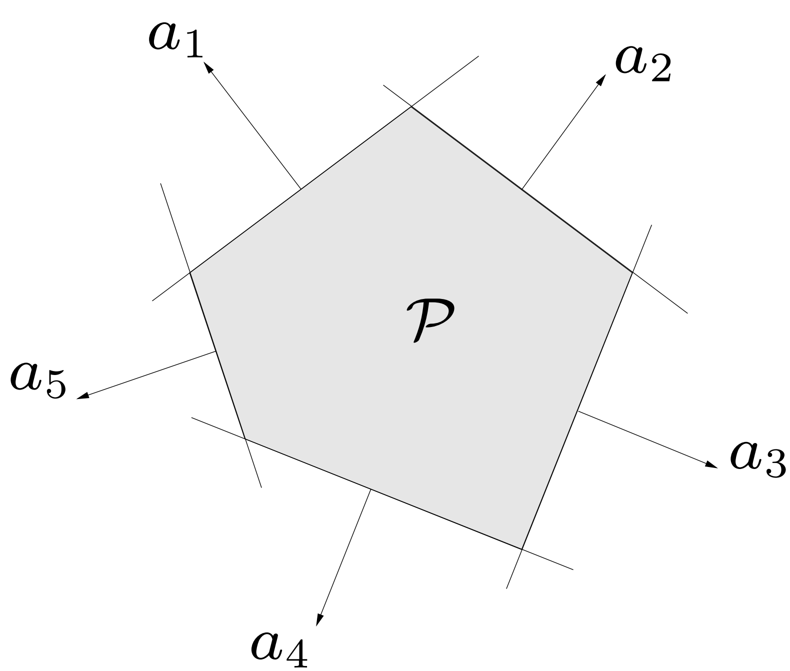
Polyhedron

A solution set of finitely many linear inequalities and equalities

$$Ax \preceq b, \qquad Cx = d$$

$x, y \in \mathbb{R}^n,\ x \preceq y \Rightarrow x_i \leq y_i\ \forall i$

($A \in \mathbf{R}^{m \times n}$, $C \in \mathbf{R}^{p \times n}$, $\preceq$ is component-wise inequality)



5 halfspace's
0 hyperplane's.

a polyhedron is an intersection of a finite number of halfspaces and hyperplanes

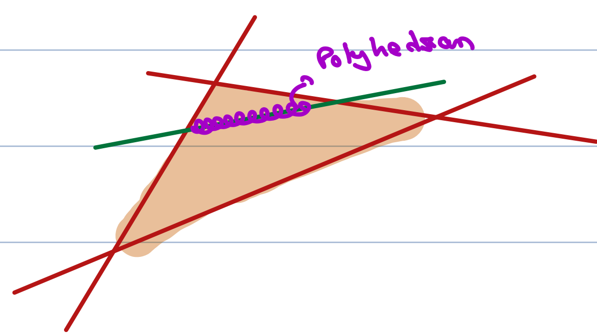
$x, y \in \mathbb{R}^n$, $x \preceq y \Rightarrow x_i \leq y_i \ \forall i$

$Ax \preceq b$ (system of linear inequalities)

$$\begin{bmatrix} - & a_1^T & - \\ - & a_2^T & - \\ & \vdots & \\ - & a_m^T & - \end{bmatrix} \begin{bmatrix} x \end{bmatrix} \preceq \begin{bmatrix} b \end{bmatrix}$$

$Cx = d$ system of linear equalities.

a Polyhedron is an intersection of a finite number of halfspace & hyper plane's.



3 – halfspaces
1 – hyperplane
– intersection.

Are Polyhedron convex? Yes
Affine? NO

Positive semidefinite cone

notation:

- $\mathbf{S}^n$ is set of symmetric $n \times n$ matrices
- $\mathbf{S}^n_+ = \{X \in \mathbf{S}^n \mid X \succeq 0\}$: positive semidefinite $n \times n$ matrices

$$X \in \mathbf{S}^n_+ \quad \Longleftrightarrow \quad z^T X z \geq 0 \text{ for all } z$$

what is PSD matrix?

X is PSD $\Longleftrightarrow$ $X \in S^n$, $z^T X z \geq 0$
$\forall z \in \mathbb{R}^n$

S^n_+ is the set of all PSD matrices

S^n_{++} is the set of all PD matrices

why S^n_+ is a convex?

$x_1, x_2 \in S^n_+ \Rightarrow z^T X_1 z \geq 0$ $\quad \forall z \in \mathbb{R}^n$
$z^T X_2 z \geq 0$

$z^T X_1 z \geq 0 \quad \times \theta$

here $\theta \geq 0$

$z^T X_2 z \geq 0 \quad \times (1-\theta)$

$\Rightarrow \theta z^T x_1 z + (1-\theta) z^T x_2 z \geq 0$

$\Rightarrow z^T (\theta x_1 + (1-\theta) x_2) z \geq 0$

$\Rightarrow \theta x_1 + (1-\theta) x_2 \in S^n_+$

$x_1, x_2 \in S^n_+$, then for $\theta \geq 0$ $\theta x_1 + (1-\theta) x_2 \in S^n_+$

$\Downarrow$

S^n_+ is convex

is S^n_+ is cone?

if $X \in S^n_+ \Rightarrow z^T X z \geq 0$

now for $\theta \geq 0$

$$\theta z^T X z \geq 0$$

$$\Rightarrow z^T (\theta X) z \geq 0$$

$$\Rightarrow \theta X \in S^n_+$$

$x \in S^n_+$ & $\forall \theta \geq 0$ $\theta x \in S^n_+$

$\Downarrow$

S^n_+ is cone

Positive semidefinite cone

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- $\mathbf{S}^n_{++} = \{X \in \mathbf{S}^n \mid X \succ 0\}$: positive definite $n \times n$ matrices

Positive semidefinite cone

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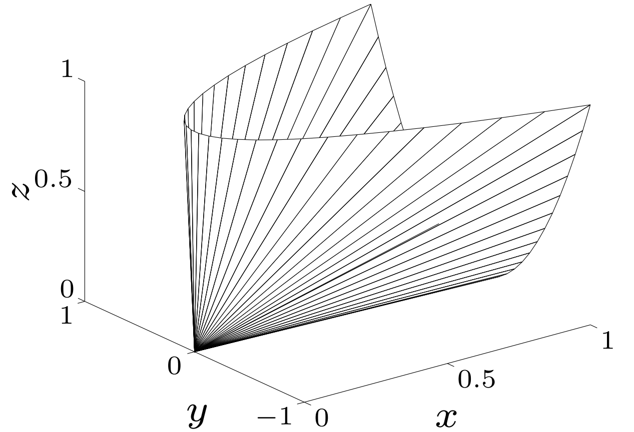
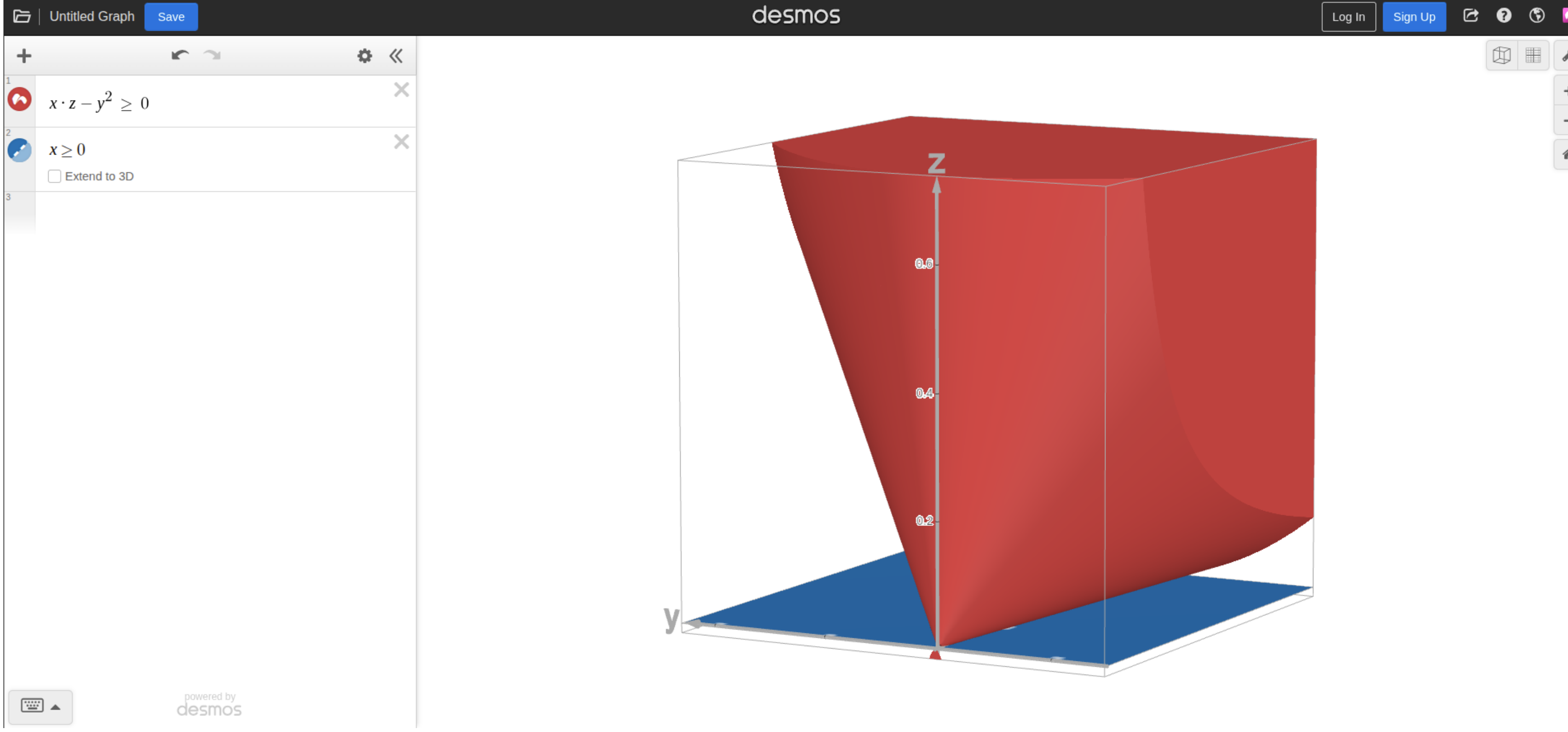
- $\mathbf{S}^n_{++} = \{X \in \mathbf{S}^n \mid X \succ 0\}$: positive definite $n \times n$ matrices

$\mathbf{S}^n_+$ is a convex cone

example: $\begin{bmatrix} x & y \\ y & z \end{bmatrix} \in \mathbf{S}^2_+$

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$x \cdot z - y^2 \geq 0$
$x \geq 0$
Extend to 3D
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Operations that preserve convexity

How to show the convexity of C?

1. apply definition

$$x_1, x_2 \in C, \quad 0 \le \theta \le 1 \quad \Longrightarrow \quad \theta x_1 + (1-\theta)x_2 \in C$$

2. show that C is obtained from simple convex sets (hyperplanes, halfspaces, ...) by operations that preserve convexity
   - Cartesian product
     ($S_1 \times S_2 = \{(x_1, x_2) \mid x_1 \in S_1,\ x_2 \in S_2\}$; if S_1 and S_2 are convex, so is $S_1 \times S_2$)
   - intersection
   - affine functions
   - perspective function
   - linear-fractional functions

So far the way we proved convexity of a set is basically by using the definition

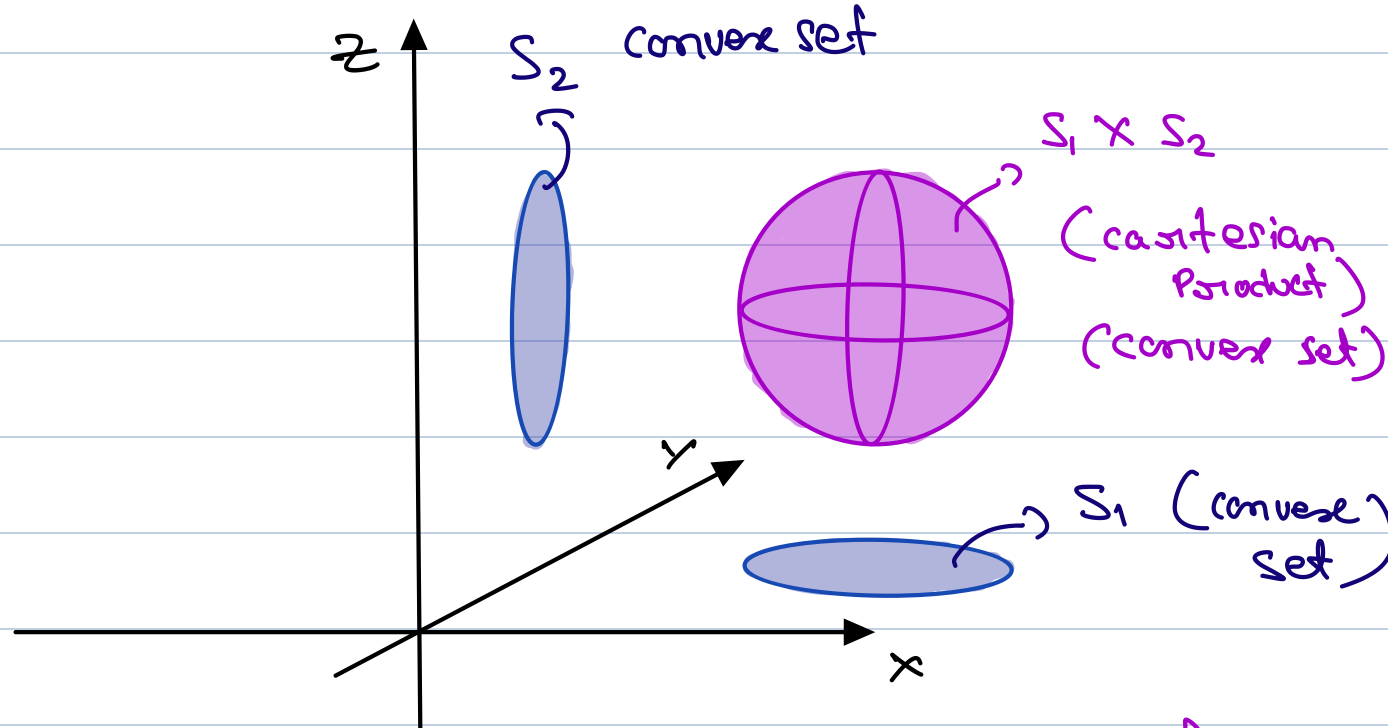
$x_1, x_2 \in C, \quad 0 \leq \theta \leq 1 \Rightarrow \theta x_1 + (1-\theta) x_2 \in C$

Pick two point's and prove $\theta x_1 + (1-\theta) x_2 \in C$

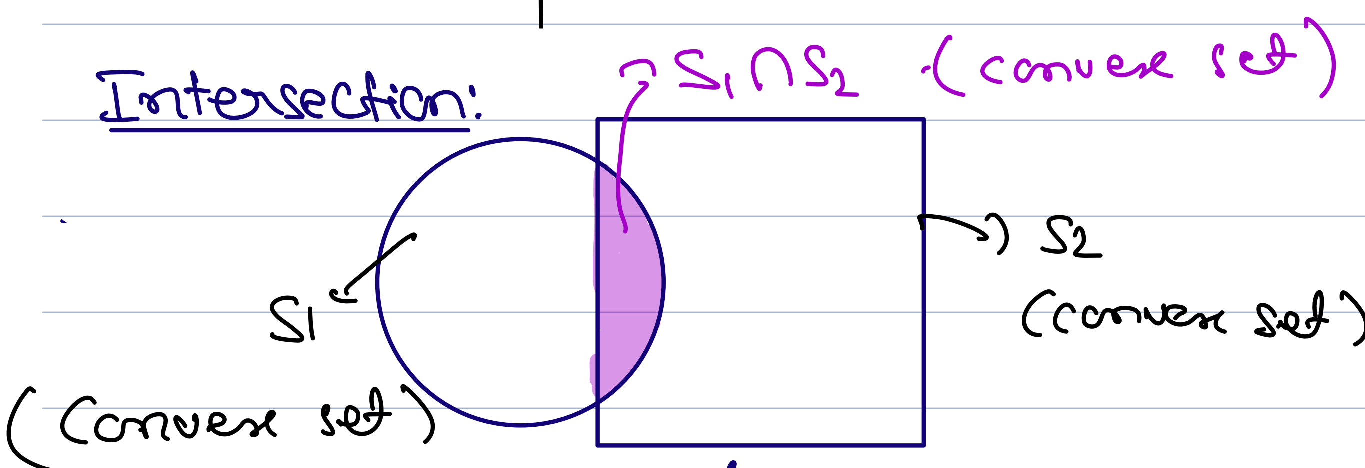
turn's out we can always use the definition, but sometime's it can become very complicated. The reason for that is sometime's our set is very complex.

Turn's out there are some operation's that preserve convexity.

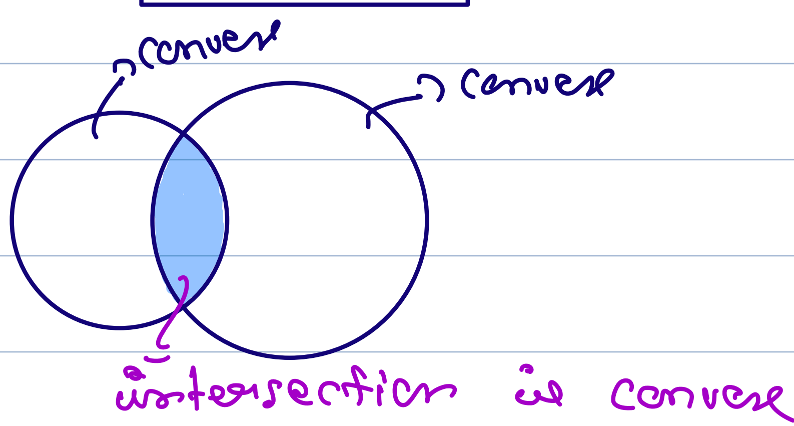
Cartesian Product:

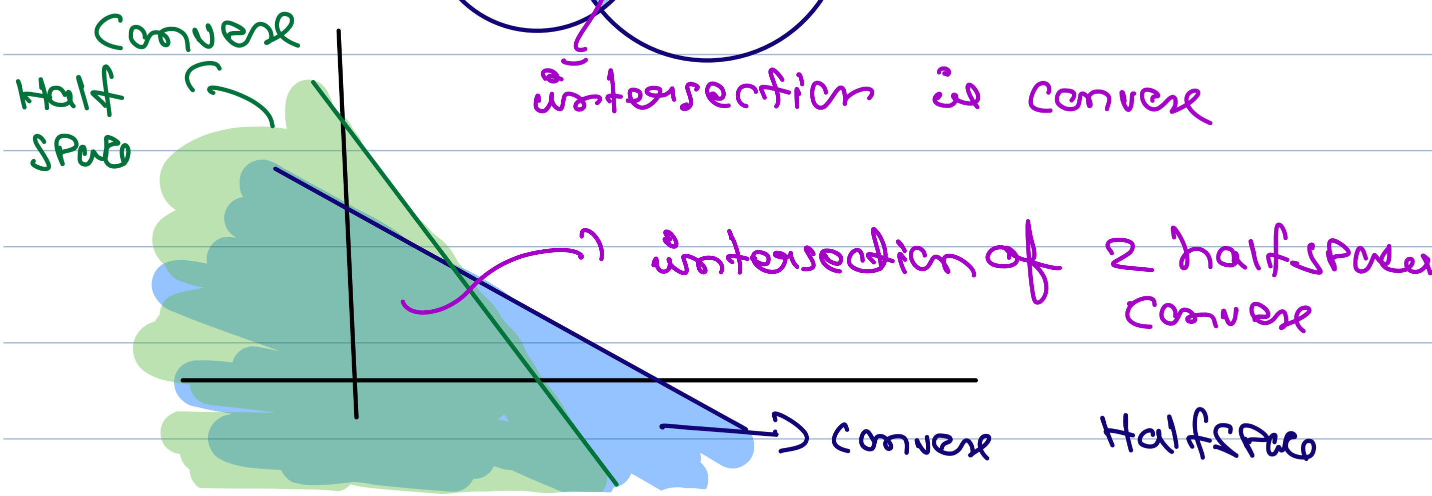


Intersection:



Ex:





Intersection

If two sets S_1 and S_2 are convex, so is their intersection $S_1 \cap S_2$.

The claim is S_1, S_2 are convex then $S_1 \cap S_2$ convex

Proof: (directly use the definition)

suppose $x_1 \in S_1 \cap S_2$ $x_2 \in S_1 \cap S_2$. ooo

goal is to show $\theta x_1 + (1-\theta) x_2 \in S_1 \cap S_2$

$$\left.\begin{array}{l} x_1, x_2 \in S_1 \Rightarrow \theta x_1 + (1-\theta) x_2 \in S_1 \\ x_1, x_2 \in S_2 \Rightarrow \theta x_1 + (1-\theta) x_2 \in S_2 \end{array}\right\} \Rightarrow \theta x_1 + (1-\theta) x_2 \in S_1 \cap S_2$$

using induction, k convex set's intersection is also convex.

PSD cone

$S_1, S_2, \ldots S_k$ convex $\Rightarrow$ $\bigcap_{i=1}^{k} S_i$ is convex.

(Application of Intersection)

$$\mathbf{S}_{+}^{n} = \{X : X \text{ is symmetric}, z^{\top} X z \geq 0, \forall z\}$$ PSD cone

Proof:

1. fix z (pick one particular z, & look at the set)

2. $C = \{x : x \in S_{+}^{n}, z^{T} x z \geq$ for some $z\}$

$$Z^T X Z = \sum Z_i Z_j X_{ij}$$

Our claim is that this is a linear function of X. why Because it's Just the Summation of different element's of X, multiplied by some fixed coefficient $Z_i Z_j$.

$$\Rightarrow \quad Z^T X Z = \sum Z_i Z_j X_{ij} \rightarrow \text{Linear w.r.t } X$$

then $Z^T X Z \geq 0$ is a half space

The set S_+^n is the intersection of Halfspace's Like above ($Z^T X Z \geq 0$ for fixed Z) for infinitely many Z. for every Z we get an different inequality, Linear in X, Halfspace. and we are looking for set of all x's that satisfy there all of these infinitely (for every Z) inequalities. each one of them is Halfspace

Half space is convex, the intersection of Half space is convex.

$$\Rightarrow \quad S^n_+ = \left\{ X : z_1^T X z_1 \geq 0,\ z_2^T X z_2 \geq 0,\ z_3^T X z_3 \geq 0, \dots \right\}$$

intersection of Bunch of Half spaces.

* But we have another set of constraints that is to impose symmetric property of X.

* X is symmetric $\Rightarrow \underbrace{x_{ij} = x_{ji}}$

This is a hyperplane

Because $\Rightarrow x_{ij} - x_{ji} = 0$

(linear equality)

therefore $$S^n_+ = \left\{ X : \begin{array}{l} z_1^T X z_1 \geq 0 \\ z_2^T X z_2 \geq 0 \\ \vdots \\ x_{ij} - x_{ji} = 0 \quad \forall i j \end{array} \right\}$$

S^n_+ is the intersection of infinitely many Halfspaces, & finite number of hyperplane. $\Rightarrow$ convex

* The intersection can be with finite set's or countably infinite sets.

i.e we do not require k to be finite
K can be countable infinite.

* But sometimes infinity is not a good thing. for example we defined "Polyhedron" for finite number of intersection of hyperplane, & Halfspaces.

* if we lift that restriction (finite restriction) then PSD cone can be treated as Polyhedron.

Affine function

suppose function $f : \mathbf{R}^n \to \mathbf{R}^m$ is affine ($f(x) = Ax + b$ with $A \in \mathbf{R}^{m \times n}$, $b \in \mathbf{R}^m$)

if S is a convex set, then the Image of that convex set under an affine function is convex.

Affine function

suppose function $f : \mathbf{R}^n \to \mathbf{R}^m$ is affine ($f(x) = Ax + b$ with $A \in \mathbf{R}^{m \times n}$, $b \in \mathbf{R}^m$)

- the image of a convex set under f is convex

$$S \subseteq \mathbf{R}^n \text{ convex} \quad \Longrightarrow \quad f(S) = \{f(x) \mid x \in S\} \subseteq \mathbf{R}^m \text{ convex}$$

  - **examples:** scaling, translation, projection, sum of sets

S is convex

scaling : $\alpha S = \{\alpha x : x \in S\}$

$\Rightarrow \quad f(x) = \underbrace{\alpha I}_{A} x$ (Affine function)

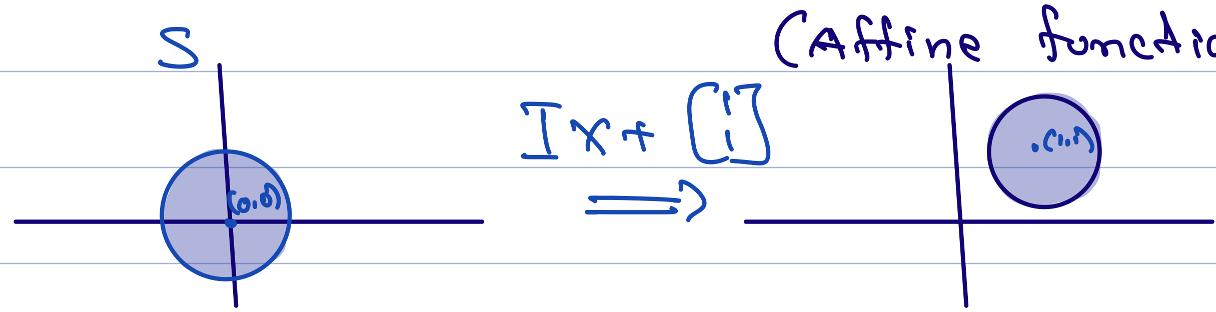
Translation: $S+a = \{x+a : x \in S\}$

$\Downarrow$

$f(x) = Ix + a$

(Affine function)

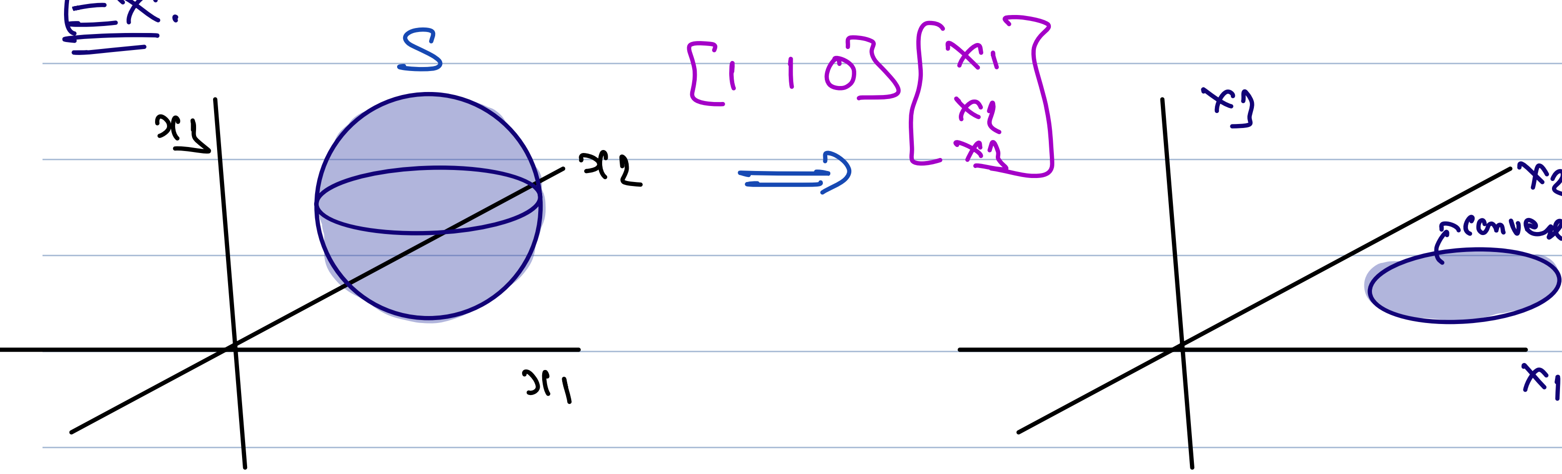
Ex:



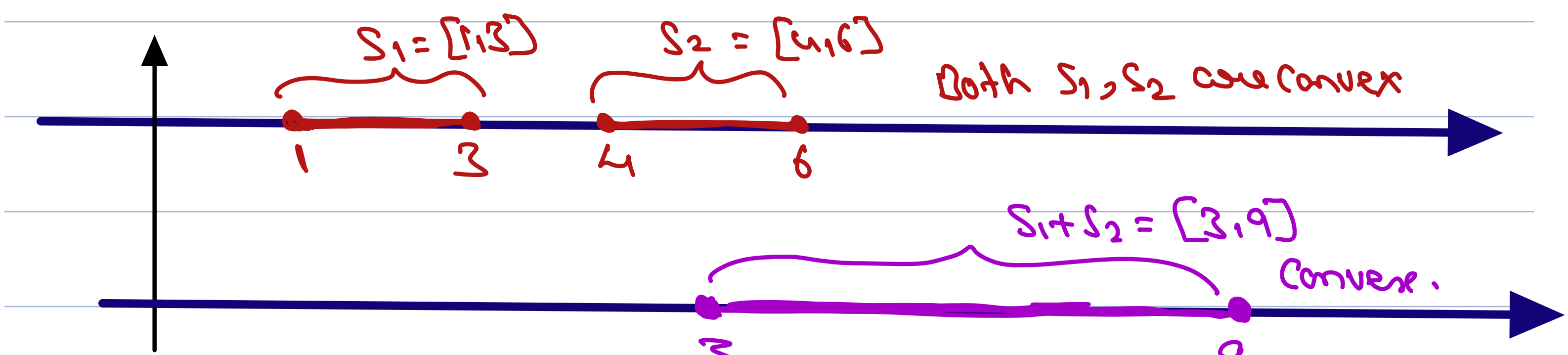
Projection: $T = \{x_1 : (x_1, x_2) \in S\}$

$$f(x) = \underbrace{\begin{bmatrix} I & 0 \end{bmatrix}}_{A} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Ex:



* **Sum's of sets** : $S_1 + S_2 = \{x_1 + x_2 : x_1 \in S_1, x_2 \in S_2\}$



Affine function

suppose function $f : \mathbf{R}^n \to \mathbf{R}^m$ is affine ($f(x) = Ax + b$ with $A \in \mathbf{R}^{m \times n}$, $b \in \mathbf{R}^m$)

- the image of a convex set under f is convex

$$S \subseteq \mathbf{R}^n \text{ convex} \quad \Longrightarrow \quad f(S) = \{f(x) \mid x \in S\} \subseteq \mathbf{R}^m \text{ convex}$$

  - **examples:** scaling, translation, projection, sum of sets

- the inverse image $f^{-1}(C)$:

$$f^{-1}(C) = \{x \in \mathbf{R}^n \mid f(x) \in C\}$$

  the inverse image of a convex set under f is convex

$$C \subseteq \mathbf{R}^m \text{ convex} \quad \Longrightarrow \quad f^{-1}(C) \text{ convex}$$

if S is convex and f is Affine function

then we want to show

$$f(S) = \{ f(x) : x \in S \} \text{ is}$$

also convex.

<u>Proof:</u> $x_1, x_2 \in S$

$$\Rightarrow f(x_1) \in S$$
$$f(x_2) \in S$$

Now consider $\theta f(x_1) + (1-\theta) f(x_2)$

$$= \theta(Ax_1 + b) + (1-\theta)(Ax_2 + b)$$

$$= A(\theta x_1 + (1-\theta)x_2) + b$$
$$\in f(S)$$

Hence Affine transformation is convex.

Examples

- **example:** solution set of linear matrix inequality
  $\{x \mid x_1 A_1 + \cdots + x_m A_m \preceq B\}$ (with $A_i, B \in \mathbf{S}^n$)